

Multiple Choice (10 marks)

1. Statement 1| ImageNet has images of various resolutions. Statement 2| Caltech-101 has more images than ImageNet.
 - (a) True, True
 - (b) False, False
 - (c) True, False
 - (d) False, True
2. Statement 1| Word 2 Vec parameters were not initialized using a Restricted Boltzman Machine. Statement 2| The tanh function is a nonlinear activation function.
 - (a) True, True
 - (b) False, False
 - (c) True, False
 - (d) False, True
3. What is the dimensionality of the null space of the following matrix? $A = \begin{bmatrix} 3, & 2, & -9 \\ -6, & -4, & 18 \\ 12, & 8, & -36 \end{bmatrix}$
 - (a) 0
 - (b) 1
 - (c) 2
 - (d) 3
4. After applying a regularization penalty in linear regression, you find that some of the coefficients of w are zeroed out. Which of the following penalties might have been used?
 - (a) L_0 norm
 - (b) L_1 norm
 - (c) L_2 norm
 - (d) either (a) or (b)
5. Averaging the output of multiple decision trees helps __.
 - (a) Increase bias
 - (b) Decrease bias
 - (c) Increase variance
 - (d) Decrease variance

True / False (10 marks)

Answer True or False

6. True or False: Residual connections are commonly utilized in architectures such as ResNets and Transformers to facilitate training deep networks.
7. True or False: $P(A, B, C)$ is equal to $P(A | B) * P(B | C) * P(C | A)$ for Boolean random variables A, B, and C.
8. True or False: Underfitting occurs when a model fails to capture the underlying patterns in the training data, resulting in poor performance on both training and new data.
9. True or False: The method used to learn the weights, whether matrix inversion or gradient descent, most critically affects the trade-off between underfitting and overfitting.
10. True or False: Clustering is effectively used for detecting fraudulent credit card transactions by grouping similar transaction patterns.

Long Form Response (20 marks)

Answer each question with a clear and complete explanation.

11. Discuss the validity of the statements regarding entropy in joint distributions and the effects of moralization on directed graphs. Analyze their implications in machine learning contexts.
12. Discuss the assumptions of the Naive Bayes algorithm, particularly focusing on the implications of attribute independence given the class value, and analyze its impact on model performance.

End of Paper